

ARGOS Weekly Update

TETHER: Causal Temporal Refinement for Frozen Surgical Stereo

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The Setting

The question

Can *one* causal temporal module be trained once and attached, unchanged, to heterogeneous **frozen** stereo estimators?

Constraints we accept

- No backbone internals.
- No future frames.
- Backbone weights never touched.

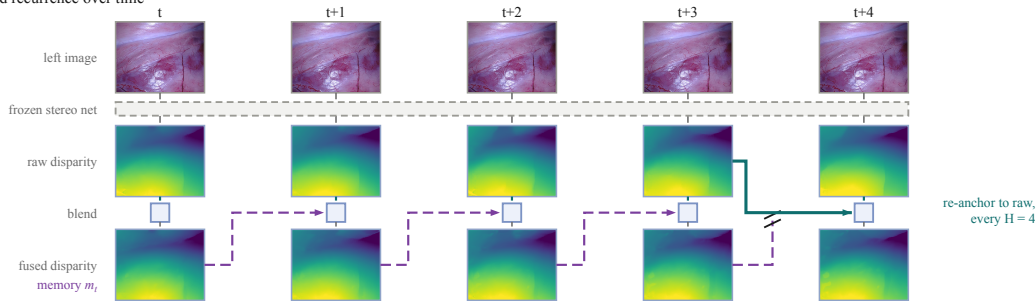
The module

177,338 parameters.

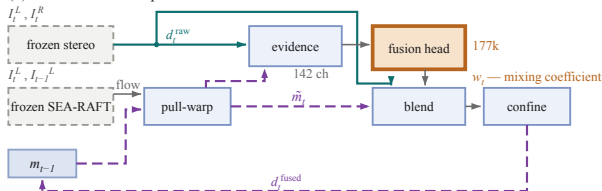
One scalar per pixel: how much to trust the past.

Architecture Overview

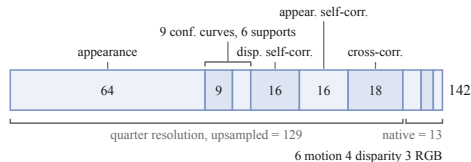
(a) bounded recurrence over time



(b) inside one time step



(c) evidence: what the 142 channels are



Grey is frozen. **One orange block is trained.** Memory is re-anchored to raw geometry every $H=4$ frames. $3/5$

It Transfers

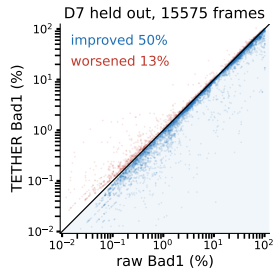
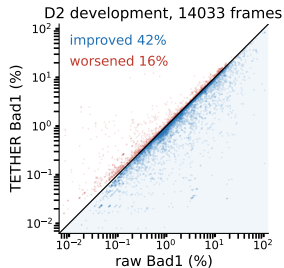
Mean EPE reduction

Backbone →	seen	unseen
SCARED-C D7	4.53%	3.70%
DRENDS (OOD)	5.31%	4.84%

One frozen checkpoint everywhere. Bottom-right is the cell usually left unevaluated.

Against a published stabiliser

Method	Causal	EPE	Bad1
Raw	—	.3029	2.09%
BiDAStabilizer	no	.3021	1.99%
TETHER	yes	.2890	1.83%



Every scored frame, all five backbones, no selection.
Below the diagonal is better.

Status

Paper

Eight pages, two figures, four tables. The head-to-head comparison, the three-seed study and the held-out transfer matrix are in.

Running now

Architecture ablations, including the one that tests whether the convex constraint is what makes the head transfer.

Next

Final touches, then submission.